# The Operator That Almost Exists: Eight New Campaigns and the Measured Boundary of the Hilbert–Pólya Program (v2.0)

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`Hilbert Polya Bridge v2.0.0`

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**Abstract**

Version 1.0 of the Hilbert–Pólya Bridge measured the boundary between the graph ($\xi$ on the critical line) and any lattice operator through three projections of one gap (rank, prime staircase, heat trace) with positive controls. Version 2.0 executes the v1 roadmap and adds eight campaigns (C9–C16) with eight theorems (T11–T18). The headline results: **(T11)** the heat trace with the Guinand–Weil kernel forces the spectrum to be the zero multiset, so a prime-weight-in-trace primitive exists and can be made arbitrarily nonlocal ($r_{H7} = 1.000$, $r_{EF} = 0.999$) — the measured H2 gap is a wall of *locality*; **(T13)** the first arithmetic-to-spectrum bridge inside the lattice: the $\alpha = 1/3$ zero modes satisfy the counting law $N_0 = 2L - 6\,[L \equiv 0 \pmod{12}]$ with Dirac index $-2$ on the $L \equiv 0 \pmod{12}$ subfamily, verified through $L = 48$; **(T14)** the verified-RH band is extended to $T = 5000$ ($N = 4520 = N_{\text{strip}}$, $S_{\max} = 0.91$); **(T17)** the trace formula transfers to Dirichlet $L$-functions with the twisted weight $\chi(n)\Lambda(n)/\sqrt{n}$ verified to $|\Delta K| \le 9.6 \cdot 10^{-6}$, and the gates stay in the null zone against $L$-zeros — the H2 gap is *universal* across the class. Blind gate families (T15), a 121-configuration anomaly taxonomy (T16), and a quantum-information layer (T18) confirm the demarcation. The package is fully self-contained: the zeta zeros are computed and committed internally (no external data files), verification is 18/18 IVP + 21/21 pytest, and an open challenge with a public scoring protocol is attached.

## 1 Introduction

The Hilbert–Pólya program asks whether the nontrivial zeros of $\zeta$ are the spectrum of a self-adjoint operator. The v1 package (C1–C8, T1–T10, 87 gates) answered five questions by measurement, with positive controls on machine precision. The roadmap named four directions able to move the main gap: nonlocal trace constructions (the T6 lesson), stitching the statistical and coding branches, a number-theoretic explanation of the $\alpha = 1/3$ zero modes, and extending the verified band. v2.0 executes all of them (C9, C10, C11, C12), adds blind campaigns and an anomaly hunt (C13, C14), transfers the pipeline to Dirichlet $L$-functions (C15), and adds a quantum-information layer (C16).

## 2 C9/T11: the nonlocal trace primitive

**Theorem 1** (Heat-trace uniqueness)**.** *Let $H$ be self-adjoint with discrete spectrum $\{\lambda_n\} \subset (0,\infty)$. If*

$$\operatorname{Tr} e^{-xH} \;=\; \frac{1}{\pi}\int_0^\infty Q(a)\sin(ax)\,da \quad \textit{for all } x>0, \qquad Q(a)=\sum_{\gamma>0}\frac{2a}{a^2+\gamma^2},$$

*then* $\operatorname{spec} H = \{\pm\gamma_n\}$ *with multiplicity.*

*Proof.* Both sides are Laplace–Stieltjes transforms of measures on $(0,\infty)$; by uniqueness the measure is determined by the trace. Since $\frac{1}{\pi}\int_0^\infty \frac{2a\,\sin(ax)}{a^2+\gamma^2}da = e^{-x\gamma}$, the measure is the counting measure of the zero set. □

**Corollary 1.** *The prime weight in the trace is the spectral measure itself: any operator with such a trace is isospectral to $H_\xi$. Nonlocality is achievable arbitrarily (dense orthogonal conjugations), locality is not.*

Measured family: $H_\xi = \operatorname{diag}(\pm\gamma)$ (control); DCT and QR twins $O\operatorname{diag}(\gamma)\,O^T$ (dense, isospectral to $7\cdot10^{-13}$): $r_{H7} = 1.000$, $r_{EF}^{(\text{fluct})} = 0.999$; truncated Berry–Keating $xp$: $r_{H7} = 0.047$ (null); the multiplicative kernel $\Lambda(\operatorname{lcm}(m,n))/\sqrt{mn}$: degenerate spectrum (an arithmetic signature), $r_{EF} = -0.96$. The T6 machine re-verifies on the internal zero cache to $|\Delta K| \le 1.8\cdot10^{-6}$.

## 3 C10/T12: hard co-location and its limit

Any phase system with $\Phi_{pq} \equiv \Phi_p + \Phi_q \pmod{2\pi}$ is a homomorphism from the free abelian group on the primes; its normal form is $\Phi_p = 2\pi\{\beta \ln p\}$. The $\{0,\pi\}$-valued case ($\Phi_p = \pi\chi_4(p)$) is *gauge-trivial* in the lattice overlay channel: $e^{\pm i\pi} = -1$ flips every $y$-bond and $\psi \to (-1)^y\psi$ removes it (spectra coincide to $8.4\cdot10^{-15}$). With the site-linked overlay the hard co-location holds at machine zero ($4.4\cdot10^{-16}$) for every $\beta$, yet the spectrum does not see it: $r_{H7} \le 0.092$ over an 11-point $\beta$-grid (Bonferroni 0.20), GUE class unchanged. Universality erases even hard codings.

## 4 C11/T13: the arithmetic bridge at $\alpha = 1/3$

For the Kähler–Dirac (staggered) operator in the $\gamma_5$-Hermitian form $H = \varepsilon D + m$ with Landau flux $\alpha = 1/3$ on the torus ($\alpha L \in \mathbb{Z}$ forces $L \equiv 0 \pmod 3$), the full scan $L = 6,\dots,48$ gives the counting law

$$N_0(L) \;=\; 2L \;-\; 6\,\big[L \equiv 0 \pmod{12}\big] \quad \text{for } L \equiv 0 \pmod 6, \qquad N_0 = 0 \text{ otherwise},$$

with Dirac index $\operatorname{ind} = \sum_{\text{zero modes}} \operatorname{sign}(\varepsilon) = -2\,[L \equiv 0 \pmod{12}]$. The decisive checks: $L = 36 \to 66$, $L = 48 \to 90$ (both exact); open boundaries give one edge zero mode with index $-1$ for odd $L \equiv 0 \pmod 3$; the $\alpha = 1/2$ control gives none. The v1 observation «tori $L \equiv 0 \pmod 4$» described only the index $-2$ subfamily: $L = 18$ ($\equiv 2 \pmod 4$) also carries 36 modes. This is the first arithmetic-to-spectrum bridge of the program; proving the law through the Kähler–Dirac index is the v3 goal.

## 5 C12/T14: the band to $T = 5000$

With $Z(t) = \texttt{mp.siegelz}$ (bridge to $e^{i\theta}\zeta$: max $|\Delta| = 0$ at six heights), adaptive sign-change counting (coarse grid 0.15, small-$|Z|$ refinement, 36-iteration bisection) and Turing-monitored $S(T)$, the count gives $N_{\text{online}}(5000) = 4520$ against $\bar{N}(5000) = 4520.08$, $S_{\max} = 0.91 \leq 2$; three full-strip argument-principle windows match (11=11, 10=10, 5=5); the v1 anchors $(0, 3, 10, 29, 79, 649)$ reproduce bit-wise. Gaps below 5000: min 0.0908, mean 1.103.

## 6 C13/T15 + C14/T16: blind families and the attractor taxonomy

Five families absent from the v1 handle space (imaginary $x$-hoppings; golden-ratio quasiperiodic vortices; vortex dimers; power-law hoppings $r^{-2}$; row flux dithering) produce 20 gates: all in the null zone ($|r_{H7}| \leq 0.040$, $\min p = 0.29$) with the pipeline control $PC0' = 1.000$. A 121-configuration scan yields the honest taxonomy 97 GUE / 18 integrable ($N_v$=0, Poisson) / 6 intermediate (weak-vortex $N_v \leq 8$, $q$=0.5, $\langle r \rangle = 0.47$–$0.53$); no rank outliers ($|r_{H7}| \leq 0.153$). The GUE attractor is global for vortex-carrying configurations, with a measurable basin structure.

## 7 C15/T17: Dirichlet $L$-functions — the gap is universal

For the quadratic characters $\chi_5$ (even) and $\chi_7$ (odd): $Z_\chi(t) = e^{i\theta_\chi(t)} L(\frac{1}{2}+it, \chi)$ is real to $5 \cdot 10^{-35}$; bisection finds 68 and 74 zeros $\leq 120$ against smooth counts 67.8 and 74.5; the twisted staircase $\psi_\chi(x) = \sum_{n \leq x} \chi(n)\Lambda(n)$ matches the zero reference at 0.902/0.949.

**Theorem 2** (Twisted trace formula)**.** $K_\chi(x) = \sum_\gamma e^{-x\gamma_\chi} = \frac{1}{\pi} \int_0^\infty Q_\chi(a) \sin(ax)\, da$*, where* $Q_\chi(a) = \frac{\Lambda'}{\Lambda}(\frac{1}{2}+a, \chi) - B_\chi - C_0^{(\chi)}$ *carries the twisted weight* $\chi(n)\Lambda(n)/\sqrt{n}$*, and* $C_0^{(\chi)} = \frac{\Lambda'}{\Lambda}(\frac{1}{2}) - B_\chi$*.*

Verified to $|\Delta K| \leq 9.6 \cdot 10^{-6}$ ($\chi_5$) and $6.1 \cdot 10^{-6}$ ($\chi_7$). The same lattice gates against $L$-zeros: $r_{H7} = 0.035$–$0.164$, the null zone. The H2 gap is a class property, not a zeta defect.

## 8 C16/T18: the quantum-information layer

Fermionic entanglement of the Slater Fermi sea (correlation-matrix protocol): the gate's half-system entropy $S_A \approx 12$–$16$ nats against the Haar baseline 160–220 ($z > 100$) — locality suppresses the volume-law entanglement, the quantum echo of C9/T11. The mutual information profile carries no zero signature ($-0.018$ vs $-0.121$ random). Spectral-form-factor slopes on unfolded 200-level windows are noisy; the gate matches GUE within the spread.

## 9 Honesty ledger (found-and-fixed, v2)

(i) the shipped $B_{\text{Hadamard}}$ constant used $\ln 2\pi$ instead of $\ln 4\pi$ (committed v1 verdicts unaffected); (ii) the v1 T9 observation refined to the full T13 law; (iii) $\{0, \pi\}$-valued codings are gauge-trivial (T12 corollary); (iv) the $r_{EF}$ metric is taken on detrended fluctuations (raw correlation catches common trends); plus the T6-check fixes ($n$ vs $\ln n$; the $a = 0.5$ pole excluded from the spline grid).

## 10 Reproducibility and the open challenge

The package uses no external data files: the zeta zeros are computed by mpmath internally and committed (800 ordinates to $\gamma = 1183.7$, bit-reproducible). Verification: IVP01–IVP18 (18/18 PASS) and pytest (21/21). An open challenge is attached: demonstrate $r_{H7} \geq 0.7$ under the v2 protocol with passing positive controls and an intact GUE class; a blind scorer (`challenge/score_submission.py`) rebuilds the gate from the submitted spec and archives the full report.

## 11 The v2 verdict

The operator that almost exists exists exactly as far as the trace allows: with the prime weight in it (T11), with arbitrary nonlocality, with prime co-location in its phases (T12). It refuses to exist locally: none of 227 measured configurations of five families and three classes left the null zones (T15, T16). Arithmetic entered the lattice spectrum where nobody looked: in the Dirac-index zero-mode count (T13). And the gap measured on zeta is a property of the whole $L$-function class (T17). The continuation is the v3 roadmap: proving the T13 law through the index, other characters and modular forms, stitching through the index, the band to $T = 50000$, and the quantum response to the index.

## References


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